

Geriatric Anesthesia MCQ – Chapter 30

Source: Short Textbook of Anesthesia – Ajay Yadav

Section 8: Subspecialty Anesthetic Management (Chapter 30)

50 Questions • 30 Minutes

Instructions:

- Each question has 4 options (A, B, C, D).
- The correct answer is shown on the right side of each question.
- Detailed explanations with textbook page references are at the end.

Questions

Q1. What is the most common cardiac finding in old age?

- A. Systolic dysfunction
- B. Diastolic dysfunction
- C. Atrial fibrillation
- D. Congestive heart failure

Ans: B

Q2. What cardiac function is usually preserved in old age despite other cardiovascular changes?

- A. Heart rate reserve
- B. Cardiac output at rest
- C. Diastolic function
- D. Systolic function

Ans: D

Q3. What happens to blood pressure with advancing age?

- A. Decreases due to reduced cardiac output
- B. Decreases due to vasodilation
- C. Remains unchanged
- D. Blood pressure increases with age

Ans: D

Q4. Which respiratory change in old age leads to V/Q mismatch and decreased pO₂?

- A. Decreased functional residual capacity
- B. Decreased residual volume
- C. Increased respiratory muscle function
- D. Increased closing capacity

Ans: D

Q5. Which spinal deformities can cause difficulty in ventilation in geriatric patients?

- A. Kyphosis and lordosis
- B. Lordosis and scoliosis
- C. Spondylosis only
- D. Kyphosis and scoliosis

Ans: D

Q6. Why is the cough reflex decreased in old age patients?

- A. Decreased respiratory muscle function decreases the cough reflex
- B. Due to increased mucus production
- C. Due to CNS depression
- D. Due to increased airway sensitivity

Ans: A

Q7. What is the CNS consequence of loss of neurons in old age?

- A. Increased cognitive function
- B. Improved neuroplasticity
- C. Increased brain reserve
- D. Decreased brain reserve leading to dementia

Ans: D

Q8. In which age group is perioperative delirium and postoperative cognitive dysfunction (POCD) far more common? A. Pediatric patients B. Middle-aged patients C. Young adults D. Old age people	Ans: D
Q9. What thermoregulation risk is the geriatric population more prone to? A. Hypothermia B. Malignant hyperthermia C. Hyperthermia D. Hyperpyrexia	Ans: A
Q10. Is age a contraindication for surgery? A. Age is not a contraindication for any surgery B. Age > 75 requires special approval C. Yes, age > 70 is a relative contraindication D. Yes, age > 80 is contraindication	Ans: A
Q11. Which preoperative assessment is a MUST for all geriatric patients? A. Echocardiography B. Renal function tests C. Assessment of cognitive function D. PFT (Pulmonary Function Tests)	Ans: C
Q12. What is the significance of preoperative dementia in geriatric patients? A. Has no perioperative impact B. Patients with preoperative dementia have higher incidence of postoperative delirium C. Requires ECT pre-treatment D. Contraindicates surgery	Ans: B
Q13. In geriatric patients, what determines when investigations should be done preoperatively? A. All patients need a complete battery of tests B. Investigations should be guided by associated comorbidity C. Patient's weight D. Age above 65 years	Ans: B
Q14. Why must a detailed record of medication be obtained in geriatric patients preoperatively? A. To identify controlled substances B. To calculate drug costs C. Many old age people are on polypharmacy and appropriate instructions must be given D. To determine drug allergies only	Ans: C

Q15. Which type of anesthesia is PREFERRED over general anesthesia in geriatric patients? A. Regional anesthesia B. Ketamine-based anesthesia C. Total IV anesthesia (TIVA) D. Monitored anesthesia care (MAC)	Ans: A
Q16. Which benefit of regional anesthesia is related to reduced postoperative venous thrombosis? A. Faster recovery B. Reduced PONV C. Reduced drug toxicity D. Central neuraxial blocks decrease the risk of thromboembolism	Ans: D
Q17. Why may central neuraxial blocks be technically difficult in geriatric patients? A. Due to increased blood pressure B. Due to patient non-cooperation C. Due to degenerative changes in spine D. Due to osteoporosis only	Ans: C
Q18. Why do old age patients achieve a higher level of spinal and epidural block? A. Due to increased drug sensitivity only B. Due to narrowing of spaces in the spine C. Due to increased CSF volume D. Due to decreased blood pressure	Ans: B
Q19. By how much should benzodiazepine premedication requirement be reduced in geriatric patients? A. 50% B. 25% C. 75% D. 100% (completely avoided)	Ans: A
Q20. Why is the dose of ALL IV induction agents decreased in geriatric patients? A. Due to smaller blood volume B. Due to reduced weight only C. Only due to reduced cardiac output D. Increased brain sensitivity, decreased volume of distribution, decreased unbound fraction (low albumin), reduced metabolism and clearance	Ans: D
Q21. Why is Propofol preferred as an induction agent in geriatric patients? A. It has the longest duration of action B. It is cheapest C. It has antiemetic properties D. Propofol because of shorter half-life is preferred	Ans: D

Q22. What is the clinical significance of decreased pseudocholinesterase levels in geriatric patients? A. Requires avoidance of all muscle relaxants B. Prolongs the effect of suxamethonium clinically C. Theoretically prolongs suxamethonium effect but clinically no prolongation is seen D. Has no effect on any drug	Ans: C
Q23. Why may intubation and mask ventilation become difficult in geriatric patients? A. Due to increased secretions B. Due to edentulous jaw and decreased neck movements from RA and osteoarthritis C. Due to macroglossia D. Due to obesity	Ans: B
Q24. What happens to stress response during surgery and anesthesia in geriatric patients? A. Remains unchanged B. Stress response to cope up with hemodynamic changes decreases C. Only hormonal response is affected D. Increases significantly	Ans: B
Q25. By how much does the MAC of inhalational agents decrease per decade in geriatric patients? A. 1-2% B. 2-3% C. 4-6% D. 10-15%	Ans: C
Q26. Which is the preferred inhalational agent for geriatric patients and why? A. Isoflurane - due to low cost B. Desflurane - because of rapid emergence and least metabolism C. Sevoflurane - due to non-pungency D. Halothane - due to cardiac stability	Ans: B
Q27. By how much should opioid doses be reduced in geriatric patients? A. No dose change required B. Reduced by half (50%) C. 25% D. 75%	Ans: B
Q28. Which is the most preferred opioid for geriatric patients? A. Remifentanyl B. Morphine C. Fentanyl D. Meperidine	Ans: A

Q29. What property of Atracurium and Cis-atracurium makes them the muscle relaxants of choice for old age patients? A. Plasma cholinesterase metabolism B. Renal elimination C. Hepatic metabolism D. Hoffman degradation	Ans: D
Q30. Why can decreased cardiac output delay the onset of muscle relaxants in geriatric patients? A. Due to larger volume of distribution B. Decreased cardiac output delays drug delivery to the neuromuscular junction C. Due to increased receptor density D. Due to increased plasma protein binding	Ans: B
Q31. What change in pain perception occurs in the postoperative period in geriatric patients? A. Studies show age-related decrease in pain perception B. Increased pain sensitivity C. Hyperpathia is common D. No change in pain sensitivity	Ans: A
Q32. What is the relationship between geriatric patients and opioid sensitivity postoperatively? A. They require higher opioid doses B. Sensitivity remains unchanged C. They are less sensitive to opioids D. They are oversensitive to opioids and therefore opioids should be used very carefully	Ans: D
Q33. Which postoperative complications are geriatric patients more prone to? A. Hypoglycemia and hypertension B. DVT and pulmonary embolism only C. Hyperthermia and urinary incontinence D. Hypoxia and urinary retention in postoperative period	Ans: D
Q34. What is the overall prevalence of postoperative delirium in older patients? A. 10% B. 50% C. 5% D. 25%	Ans: A
Q35. To what percentage can postoperative delirium incidence increase in patients shifted to ICU? A. 15-20% B. 30-35% C. 25% D. More than 50%	Ans: D

Q36. Which age is cited as an important predisposing factor for postoperative delirium? A. Age > 50 years B. Age > 60 years C. Age > 75 years D. Age > 65 years	Ans: D
Q37. Which type of surgery is a predisposing factor for postoperative delirium? A. Ophthalmic surgery B. Laparoscopic surgery only C. Major surgery D. Minor elective surgery	Ans: C
Q38. Which analgesic drug is listed as predisposing to postoperative delirium? A. Fentanyl B. Tramadol C. Morphine D. Meperidine (pethidine)	Ans: D
Q39. Which of the following is NOT listed as a predisposing factor for postoperative delirium? A. Sleep deprivation B. Polypharmacy C. Female gender D. Electrolyte imbalance	Ans: C
Q40. What are the short-term changes after surgery related to POCD in geriatric patients? A. Only affects patients >80 years B. Short-term (up to 3 months) changes in cognitive function well-documented in 10-15% of geriatric population after surgery C. Short-term changes in 30-40% of patients D. Permanent cognitive decline in all patients	Ans: B
Q41. What percentage of patients can have long-term (>3 months) cognitive impairment after surgery? A. 5% B. 10% C. 15% D. 1%	Ans: D
Q42. What is the relationship between POCD and the type of surgery/anesthesia? A. POCD is more common with general anesthesia B. POCD is more common with regional anesthesia C. The incidence of POCD is independent of both surgery and anesthesia D. POCD is more common with major surgery	Ans: C

Q43. What is the current treatment for POCD? A. Cognitive rehabilitation therapy B. Cholinesterase inhibitors C. At present there is no specific treatment available for POCD D. NSAID prophylaxis	Ans: C
Q44. What change in albumin levels occurs in old age and what is its pharmacological significance? A. No change in albumin levels B. Albumin decreases, leading to decreased unbound fraction of drug, decreasing dose requirements but increasing toxicity C. Albumin increases, so more drug is protein-bound D. Decreased albumin leads to decreased unbound fraction	Ans: B
Q45. What happens to the volume of distribution of drugs in geriatric patients? A. Only water-soluble drug distribution changes B. Increases for all drugs C. Decrease in volume of distribution D. Volume of distribution is unchanged	Ans: C
Q46. Why does general anesthesia have a higher incidence of postoperative delirium compared to regional anesthesia? A. Due to increased stress response under GA B. General anesthesia (compared to regional anesthesia) is an important predisposing factor for postoperative delirium C. Due to higher opioid use under GA D. Due to longer duration of surgery under GA	Ans: B
Q47. What happens to residual volume and functional residual capacity in old age due to decreased lung elasticity? A. Decreased elasticity leads to alveolar distension and increase in residual volume and functional residual capacity B. Residual volume decreases, FRC increases C. Both decrease D. Both remain unchanged	Ans: A
Q48. What happens to the adrenergic response with age and what is its clinical significance? A. Only alpha-adrenergic response is affected B. Increased adrenergic response causing hypertension C. Decreased adrenergic response, contributing to increased incidence of cardiac complications D. No change in adrenergic response	Ans: C

Q49. Which pulmonary complication associated with general anesthesia can be avoided by using regional anesthesia in geriatric patients?

- A. Aspiration pneumonitis only
- B. Pulmonary edema only
- C. Pulmonary complications associated with general anesthesia may be avoided with regional anesthesia
- D. Pneumothorax only

Ans: C

Q50. What is the key statement regarding chronological age and surgery/investigations in geriatric patients?

- A. Age > 70 requires full workup before surgery
- B. Chronological age is neither a contraindication for any surgery nor a criterion to do battery of routine investigations
- C. Chronological age is a contraindication for high-risk surgery
- D. Age > 65 requires cardiac clearance

Ans: B

Answer Key & Explanations

Q1. Answer: B — Diastolic dysfunction

Page 242: Decreased ventricular compliance leading to diastolic dysfunction is the most common finding in old age. It is also a significant determinant of morbidity.

Topic: Physiological Changes - Cardiovascular

Q2. Answer: D — Systolic function

Page 242: Systolic function usually remains preserved in old age, whereas diastolic dysfunction is the most common finding due to decreased ventricular compliance.

Topic: Physiological Changes - Cardiovascular

Q3. Answer: D — Blood pressure increases with age

Page 242: Blood pressure increases with age. This along with decreased heart rate, decreased adrenergic response, and decreased cardiac reserve increase the incidence of cardiac complications.

Topic: Physiological Changes - Cardiovascular

Q4. Answer: D — Increased closing capacity

Page 242: Increase in closing capacity increases the V/Q mismatch decreasing the pO₂. This is due to decreased elasticity leading to alveolar distension.

Topic: Physiological Changes - Respiratory

Q5. Answer: D — Kyphosis and scoliosis

Page 242: Kyphosis and scoliosis can cause difficulty in ventilation in geriatric patients.

Topic: Physiological Changes - Respiratory

Q6. Answer: A — Decreased respiratory muscle function decreases the cough reflex

Page 242: Decreased respiratory muscle function decreases the cough reflex in geriatric patients, making them more prone to aspiration pneumonia.

Topic: Physiological Changes - Respiratory

Q7. Answer: D — Decreased brain reserve leading to dementia

Page 242: Loss of neurons decreases the brain reserve leading to dementia. There also occurs increased sensitivity to anesthetic medications.

Topic: Physiological Changes - CNS

Q8. Answer: D — Old age people

Page 242: The risk of perioperative delirium and postoperative cognitive dysfunction (POCD) is far more common in old age people due to decreased brain reserve.

Topic: Physiological Changes - CNS

Q9. Answer: A — Hypothermia

Page 242: Geriatric population is more prone for hypothermia due to impaired thermoregulation mechanisms.

Topic: Physiological Changes - Thermoregulation

Q10. Answer: A — Age is not a contraindication for any surgery

Page 242: Age is not a contraindication for any surgery. Chronological age is neither a contraindication for any surgery nor a criterion to do battery of routine investigations.

Topic: Anesthetic Management - General

Q11. Answer: C — Assessment of cognitive function

Page 242: Assessment of cognitive function is must. Patients with preoperative dementia have higher incidence of postoperative delirium.

Topic: Anesthetic Management - Preoperative

Q12. Answer: B — Patients with preoperative dementia have higher incidence of postoperative delirium

Page 242: Patients with preoperative dementia have higher incidence of postoperative delirium. Moreover, cognitive assessment can be used to compare if patient develops cognitive function in the postoperative period.

Topic: Anesthetic Management - Preoperative

Q13. Answer: B — Investigations should be guided by associated comorbidity

Page 242: Age is not a criteria to do battery of routine investigations. Investigations should be guided by associated comorbidity.

Topic: Anesthetic Management - Preoperative

Q14. Answer: C — Many old age people are on polypharmacy and appropriate instructions must be given

Page 242: Many of the old age people are on polypharmacy. A detailed record of medication should be obtained and instructions are given accordingly.

Topic: Anesthetic Management - Preoperative

Q15. Answer: A — Regional anesthesia

Page 242: Regional anesthesia is preferred over general anesthesia due to increased sensitivity of brain to anesthetic agents, decreased metabolism/excretion, reduced chances of drug toxicity, decreased thromboembolism risk, fewer pulmonary complications, and less incidence of postoperative delirium.

Topic: Anesthetic Management - Choice of Anesthesia

Q16. Answer: D — Central neuraxial blocks decrease the risk of thromboembolism

Page 243: Central neuraxial blocks decrease the risk of thromboembolism. This is one reason regional anesthesia is preferred over general anesthesia in geriatric patients.

Topic: Anesthetic Management - Choice of Anesthesia

Q17. Answer: C — Due to degenerative changes in spine

Page 243: Regional anesthesia, particularly central neuraxial blocks, may be technically difficult due to degenerative changes in spine.

Topic: Anesthetic Management - Choice of Anesthesia

Q18. Answer: B — Due to narrowing of spaces in the spine

Page 243: Old age patients achieve higher level of spinal (and epidural) due to narrowing of spaces. Geriatric population is also more sensitive to the effect of local anesthetics during nerve blocks.

Topic: Anesthetic Management - Choice of Anesthesia

Q19. Answer: A — 50%

Page 243: As the requirement of benzodiazepines may be reduced by 50%, premedication with benzodiazepines should be done cautiously in geriatric patients.

Topic: Anesthetic Management - General Anesthesia

Q20. Answer: D — Increased brain sensitivity, decreased volume of distribution, decreased unbound fraction (low albumin), reduced metabolism and clearance

Page 243: Increase sensitivity of brain, decrease volume of distribution, decreased unbound fraction (due to decreased albumin levels) and reduced metabolism and clearance not only decreases the dose of all IV agents but also increases toxicity.

Topic: Anesthetic Management - General Anesthesia

Q21. Answer: D — Propofol because of shorter half-life is preferred

Page 243: Propofol because of shorter half-life is preferred. Decreased cardiac output may slow the onset with IV agents.

Topic: Anesthetic Management - General Anesthesia

Q22. Answer: C — Theoretically prolongs suxamethonium effect but clinically no prolongation is seen

Page 243: Decreased pseudocholinesterase levels due to decreased hepatic functions may theoretically prolong the effect of suxamethonium, however clinically no prolongation has been seen.

Topic: Anesthetic Management - General Anesthesia

Q23. Answer: B — Due to edentulous jaw and decreased neck movements from RA and osteoarthritis

Page 243: Intubation and mask ventilation may become difficult due to edentulous jaw and decreased neck movements (due to rheumatoid arthritis and osteoarthritis).

Topic: Anesthetic Management - General Anesthesia

Q24. Answer: B — Stress response to cope up with hemodynamic changes decreases

Page 243: Stress response to cope up with hemodynamic changes during surgery and anesthesia decreases in geriatric patients.

Topic: Anesthetic Management - General Anesthesia

Q25. Answer: C — 4-6%

Page 243: The minimum alveolar concentration (MAC) of inhalational agents decreases by 4-6% per decade in geriatric patients.

Topic: Anesthetic Management - General Anesthesia

Q26. Answer: B — Desflurane - because of rapid emergence and least metabolism

Page 243: Desflurane because of its rapid emergence and least metabolism is the preferred inhalational agent for geriatric patients.

Topic: Anesthetic Management - General Anesthesia

Q27. Answer: B — Reduced by half (50%)

Page 243: Doses of opioids should be reduced by half in geriatric patients due to age-related decrease in drug metabolism and excretion.

Topic: Anesthetic Management - General Anesthesia

Q28. Answer: A — Remifentanyl

Page 243: Remifentanyl because of its ultra-short life is the most preferred opioid for old age patients.

Topic: Anesthetic Management - General Anesthesia

Q29. Answer: D — Hoffman degradation

Page 243: Hoffman degradation makes Atracurium and Cis-atracurium to be the muscle relaxants of choice for old age patients as it is independent of hepatic and renal function.

Topic: Anesthetic Management - General Anesthesia

Q30. Answer: B — Decreased cardiac output delays drug delivery to the neuromuscular junction

Page 243: Decrease cardiac output can delay the onset of muscle relaxants in geriatric patients as the drug takes longer to reach the neuromuscular junction.

Topic: Anesthetic Management - General Anesthesia

Q31. Answer: A — Studies show age-related decrease in pain perception

Page 243: Studies have shown that there occurs age related decrease in pain perception in geriatric patients in the postoperative period.

Topic: Anesthetic Management - Postoperative

Q32. Answer: D — They are oversensitive to opioids and therefore opioids should be used very carefully

Page 243: Geriatric patients are oversensitive to opioids, therefore opioids should be used very carefully in the postoperative period.

Topic: Anesthetic Management - Postoperative

Q33. Answer: D — Hypoxia and urinary retention in postoperative period

Page 243: Geriatric population is more prone for hypoxia and urinary retention in the postoperative period.

Topic: Anesthetic Management - Postoperative

Q34. Answer: A — 10%

Page 243: The overall prevalence of postoperative delirium in older patients is estimated to be 10%.

Topic: Postoperative Delirium

Q35. Answer: D — More than 50%

Page 243: The incidence of delirium can increase to more than 50% in patients shifted to ICU.

Topic: Postoperative Delirium

Q36. Answer: D — Age > 65 years

Page 243: Important predisposing factors are age >65 years, male gender, major surgery, general anesthesia (compared to regional anesthesia), and other factors like drugs and comorbidities.

Topic: Postoperative Delirium

Q37. Answer: C — Major surgery

Page 243: Important predisposing factors include major surgery (compared to minor), general anesthesia (compared to regional anesthesia), age >65 years and male gender.

Topic: Postoperative Delirium

Q38. Answer: D — Meperidine (pethidine)

Page 243: Drugs like meperidine (pethidine) are listed as predisposing factors for postoperative delirium in geriatric patients.

Topic: Postoperative Delirium

Q39. Answer: C — Female gender

Page 243: Male gender (not female gender) is listed as a predisposing factor. Other predisposing factors include age >65 years, major surgery, GA, meperidine, cognitive impairment, dementia, comorbidities, polypharmacy, electrolyte imbalance, pain, and sleep deprivation.

Topic: Postoperative Delirium

Q40. Answer: B — Short-term (up to 3 months) changes in cognitive function well-documented in 10-15% of geriatric population after surgery

Page 243: Short-term (up to 3 months) changes in cognitive function have been very well-documented in 10-15% of geriatric population after surgery.

Topic: Postoperative Cognitive Dysfunction (POCD)

Q41. Answer: D — 1%

Page 243: Long-term (>3 months) cognitive impairment can also occur in 1% of the patients.

Topic: Postoperative Cognitive Dysfunction (POCD)

Q42. Answer: C — The incidence of POCD is independent of both surgery and anesthesia

Page 243: The incidence of postoperative cognitive dysfunction (POCD) is independent of both surgery and anesthesia. Studies have shown no difference in the incidence of POCD in minor versus major surgery and regional versus general anesthesia.

Topic: Postoperative Cognitive Dysfunction (POCD)

Q43. Answer: C — At present there is no specific treatment available for POCD

Page 243: At present there is no specific treatment available for POCD.

Topic: Postoperative Cognitive Dysfunction (POCD)

Q44. Answer: B — Albumin decreases, leading to decreased unbound fraction of drug, decreasing dose requirements but increasing toxicity

Page 243: Decreased albumin levels in old age lead to decreased unbound fraction of the drug. This along with reduced metabolism and clearance decreases the dose of all IV agents but also increases toxicity.

Topic: Anesthetic Management - Pharmacology

Q45. Answer: C — Decrease in volume of distribution

Page 243: Decrease volume of distribution is one of the reasons why dose of all IV agents decreases in geriatric patients, along with increased brain sensitivity, decreased albumin, and reduced metabolism.

Topic: Anesthetic Management - Pharmacology

Q46. Answer: B — General anesthesia (compared to regional anesthesia) is an important predisposing factor for postoperative delirium

Page 243: General anesthesia compared to regional anesthesia is listed as an important predisposing factor for postoperative delirium in geriatric patients. This is also why regional anesthesia is preferred.

Topic: Postoperative Delirium

Q47. Answer: A — Decreased elasticity leads to alveolar distension and increase in residual volume and functional residual capacity

Page 242: Decreased elasticity leads to alveolar distension and increase in residual volume and functional residual capacity in old age.

Topic: Physiological Changes - Respiratory

Q48. Answer: C — Decreased adrenergic response, contributing to increased incidence of cardiac complications

Page 242: Decrease in adrenergic response along with decreased heart rate and decreased cardiac reserve increase the incidence of cardiac complications in old age.

Topic: Physiological Changes - Cardiovascular

Q49. Answer: C — Pulmonary complications associated with general anesthesia may be avoided with regional anesthesia

Page 243: Pulmonary complications associated with general anesthesia (GA) may be avoided with the use of regional anesthesia in geriatric patients. This is one of the advantages of regional anesthesia.

Topic: Anesthetic Management - Choice of Anesthesia

Q50. Answer: B — Chronological age is neither a contraindication for any surgery nor a criterion to do battery of routine investigations

Page 244 (Key Points): Chronological age is neither a contraindication for any surgery nor a criterion to do battery of routine investigations. Investigations should be guided by associated comorbidity.

Topic: Key Points